The Klootwijk Architectural Paradigm

Version 3.0

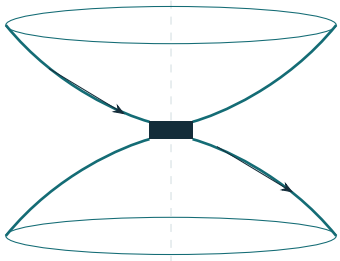
Two preparaats. Open cups. Whaletail.

Biochemical and atomic clocks with causal self-reference

One tick. One bit toggle. One π phase advance.

$$\mathcal{T}_\pi : (N, q, \Theta) \longmapsto (N+1, q \oplus 1, \Theta + \pi).$$

Section 20: convergence $<$ $\longrightarrow$ committed event $\longrightarrow$ Section 21: expansion $>$



projection $<$

$\mathcal{T}_\pi$ / Ψ_{dove}

self-reference $>$

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Based on the attached *wait I got 2 preparaats.pdf*, with the biochemical, atomic-clock and executable foundations carried forward from Version 2.0.

Architectural statement

Preparaat 1 supplies structured capacity. Preparaat 2 supplies the activating process. Their encounter evolves a conserved switch population. Clock Alpha responds; Clock Beta retains; the hourglass integrates dwell. Each π-tick changes the one-bit register, commits the resulting state, and makes that record available to the next execution interval.

The attached source makes the geometry more specific: two opposed **open**, **non-globular cups**, a central axle, two external stick-poles, and a string that supplies drive rather than defining the stored spatial route. Its final whaletail description links the flared transport geometry to the quarter-turn crosshatch and the $\mathcal{T}_{\pi}$ event. [S3, pp. 10–15]

The sequence retained from the attached source

The source first names the two preparaats, then collects conservation, interaction, dual clocks, dwell and ticks into one loop. It traces the record mechanism back to Sections 20 and 21, reads them as convergence $<$ and expansion $>$, replaces the global horseshoe interpretation with the diabolo, and develops topspin, gravity and the whaletail. V3 follows that sequence while retaining the biochemical and atomic-clock modules of V2. Sections **20 and 21 keep their original functions and numbers**. [S3, pp. 2–15; V2, §§20–21]

How the formal text is organized

Source basis identifies statements or equations present in the supplied documents. **V3 construction** identifies the explicit maps, constitutive choices and geometrical definitions introduced here. A derived identity is accompanied by its calculation or assumptions. This separates a source term, its mathematical meaning, and the equation that realizes it.

The phrases *petrification*, *cryptographic memory locking*, *topspin*, *red herring* and *whaletail* remain in the architecture. The committed payload, its digest, mechanical angular momentum, chemical bias and geometric routing are assigned separate coordinates so that these terms can participate in an executable system.

The executed V3 instance

A 48-tick reference execution integrates the two-state chemical fraction, Alpha/Beta dynamics, cyclic memory and a driven rotor. The rotor and bit select anisotropic routing weights. Those weights and the system's previously emitted records change its next held chemical rate. Every interval is counted in integer reference cycles. The actual results, model identity and event digest are included in this document and its companion files.

The source gives qualitative material roles rather than a measured species list, shell profile or torque series. The numerical coefficients in §18 and Appendix B are therefore declared instance parameters. The shell, optical, spatial transport and atomic interrogation equations remain specified modules rather than additional simulated fields.

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1 Preparaat 1 and Preparaat 2

Source basis S3, pp. 2–4; V2, §§3 and 6.

The base document names **Preparaat 1** as *Structured Capacity* and **Preparaat 2** as the *Activating/Mobilized Process*. Its page 2 associates Preparaat 1 with the named file *militair-cologne.pdf*; its page 3 associates Preparaat 2 with V2. Here those names retain the roles actually stated in the attached dialogue.

For a chosen convertible population $P_\star > 0$ and activated fraction c,

$$P_1 = P_\star(1-c), \qquad P_2 = P_\star c, \qquad 0 \le c \le 1. \tag{1.1}$$

Consequently,

$$P_1 + P_2 = P_\star, \qquad \dot{P}_1 = -P_\star \dot{c}, \qquad \dot{P}_2 = P_\star \dot{c}. \tag{1.2}$$

The pair is a capacity/activation partition of the same designated population. Scaffold sites, lipids, water, fuel and other material constituents have their own species balances; Equation (1.1) is not a conversion of one entire material class into another.

Double Hadamard interaction

The source's co-located intersection is retained as

$$\mathcal{C} = W \odot (P_1 \odot P_2), \qquad \widehat{\mathcal{C}} = W \odot [c \odot (1-c)]. \tag{1.3}$$

Here W is an equally indexed spatial or optical overlap mask. If P_1, P_2 have concentration units, $\mathcal{C}$ has concentration-squared units; a reaction law using it includes its corresponding rate coefficient. The normalized interaction $\widehat{\mathcal{C}}$ is dimensionless when W is dimensionless.

For $W \ge 0$, the scalar interaction satisfies

$$0 \le Wc(1-c) \le \frac{W}{4}, \tag{1.4}$$

because $c(1-c) = \frac{1}{4} - (c - \frac{1}{2})^2$. The maximum is at equal latent and activated fractions. This is an overlap identity, not an automatic kinetic law.

Material and geometric realizations

In the biochemical realization, $P_1 = [B_0]$ and $P_2 = [B_1]$ are two switch-state concentrations. In the transport realization, Preparaat 1 supplies the supporting channel structure and Preparaat 2 the mobilized process. V3 changes the global geometry to the source's open-cup configuration while retaining these roles. The conserved population, the geometric scaffold and the active flux are related by explicit constitutive maps rather than by equating their units.

2 The five-step loop and the one-bit π-tick

Source basis S3, pp. 4–5; the author's preceding instruction defining a tick as a bit flip over π; V2, §1.

The source's five steps become a single ordered operation: establish the preparaat partition; evaluate their masked interaction; evolve Alpha and Beta; integrate dwell and recovery; resolve the event and commit its state.

Let $N \in \mathbb{N}_0$ count events, $q \in \{0, 1\}$ be the stored bit, and Θ be the unwrapped event phase. Define

$$\boxed{\mathcal{T}_\pi(N, q, \Theta) = (N + 1, q \oplus 1, \Theta + \pi).} \tag{2.1}$$

For $N_0 = q_0 = \Theta_0 = 0$,

$$N_n = n, \qquad q_n = n \bmod 2, \qquad \Theta_n = n\pi. \tag{2.2}$$

The circular phase and its signed representation are

$$\theta_n = \Theta_n \bmod 2\pi = \pi q_n, \qquad z_n = e^{i\theta_n} = (-1)^{q_n}, \qquad z_{n+1} = -z_n. \tag{2.3}$$

Thus a tick is one transition in a one-bit register. The constant 1 in $q \oplus 1$ is the nonzero element of $\mathbb{Z}_2$; it complements the state. The 1 in $N + 1$ increments an integer count. Their shared notation does not merge those two state spaces.

A complete turn does not erase the history

Two successive ticks give

$$\mathcal{T}_\pi^2(N, q, \Theta) = (N + 2, q, \Theta + 2\pi). \tag{2.4}$$

The bit returns, but the count advances. With an append-only execution history,

$$\mathcal{H}_{n+2} = \mathcal{H}_n \| D_{n+1} \| D_{n+2}, \tag{2.5}$$

the full state is also different. Here $\|$ denotes ordered concatenation.

What the bit counts

A binary register has two alternatives. A predetermined complement changes its stored value without implying one new independent random bit. For the deterministic event rule, $H(q_{n+1} \mid q_n) = 0$. Timing, measured inputs and record contents are separate information-bearing variables. V3 retains the author's identity as an **event convention**: one tick, one complement, one encoded half-turn. [V2, §1]

> The five steps use one event boundary. The chemistry evolves during the interval; the register changes at its endpoint; the committed output becomes an input to the following interval.

3 Typed state, quarter-turn crosshatch and continuous spin

Source basis S3, pp. 11–15; V2, §§2–3.

V3 construction The continuous diabolo spin is given its own angle, distinct from the two-state crosshatch and event phase.

Use the state

$$\mathcal{S} = (x, q, N, \Theta, C_A, \mathcal{H}, \mathcal{M}), \quad x = (C, T, \ell, u_s, p, c, a, b, m, X, \psi, \omega_s). \tag{3.1}$$

C contains material species, T temperature, ℓ lipid order, u_s scaffold displacement and p pore pressure. The reduced instance selects only $(c, a, b, m, X, \psi, \omega_s)$ from these continuous variables.

Symbol	Coordinate	Unit/type
c, m, ℓ	Activated fraction, retained occupancy, order	dimensionless
a, b	Clock Alpha and Clock Beta	normalized state
X	Accumulated active dwell	s
ψ, ω_s	Mechanical rotor angle and angular speed	rad, rad s^{-1}
q, N	Bit and event count	Boolean, integer
$\Theta, \Phi_A, \vartheta_R$	Event phase, reference carrier phase, Rabi area	separate angles
$D, h_D, \mathcal{H}$	Payload, digest, ordered archive	data objects

The inherited crosshatch labels two orientations separated by 90°:

$$\varphi_q = \frac{\pi q}{2}, \qquad R_q = R_z(\varphi_q), \qquad \theta_{\text{enc}} = 2\varphi_q = \pi q \pmod{2\pi}. \tag{3.2}$$

This is the explicit bridge between a mechanical quarter-turn and the encoded π-flip. At fixed body frame, successive changes are $+\pi/2$ and $-\pi/2$. They need not equal the direction of continuous diabolo spin.

With body orientation $Q_s(t) = R_z(\psi(t))$, the laboratory transport orientation is

$$Q(t, q) = Q_s(t) R_q, \qquad \dot{\psi} = \omega_s. \tag{3.3}$$

The rotor may keep spinning positively while q alternates. A sampled anisotropic tensor can therefore change continuously through ψ and discretely through q.

Three distinct half-turns. An event-phase advance is $\Delta\Theta = \pi$. The inherited crosshatch step has magnitude $\pi/2$. An atomic population inversion requires a driven Rabi area $\vartheta_R = \pi$. These are linked by the declared maps, not by assigning one angle to every subsystem.

4 Biochemical equations and conserved moieties

Source basis S3, pp. 3–4; V2, §§6 and 9.

For the two switch states,

$$B_0 \underset{k_{10}}{\overset{k_{01}}{\rightleftharpoons}} B_1, \qquad c = \frac{[B_1]}{C_\star}, \qquad C_\star = [B_0] + [B_1]. \tag{4.1}$$

At fixed total population and first-order or chemostatted pseudo-first-order rates,

$$\dot{c} = k_{01}(1 - c) - k_{10}c, \qquad c_\infty = \frac{k_{01}}{k_{01} + k_{10}}. \tag{4.2}$$

On a held interval of length Δt the exact solution is

$$c^+ = c_\infty + (c - c_\infty)e^{-(k_{01}+k_{10})\Delta t}. \tag{4.3}$$

The coefficients of c and c_∞ form a convex combination. Nonnegative rates therefore preserve $0 \le c \le 1$ without modifying the result after the step.

Network form

For species concentrations C, stoichiometric matrix S and net channel rates r,

$$\dot{C} = Sr + s_{\text{ext}}, \qquad r_j = k_j^+ \prod_i C_i^{\nu_{ij}^-} - k_j^- \prod_i C_i^{\nu_{ij}^+}. \tag{4.4}$$

Every l satisfying $l^T S = 0$ gives

$$\frac{\mathrm{d}}{\mathrm{d}t}(l^T C) = l^T s_{\text{ext}}. \tag{4.5}$$

Thus a closed switch conserves its selected moiety. With boundary exchange, conservation is expressed by the corresponding integral balance, not by suppressing the exchange.

Transport through Preparaat 1

Let ϕ_p denote porosity, j Darcy volume flux and D_i a diffusion/dispersion tensor. Species conservation per bulk volume is

$$\partial_t(\phi_p C_i) + \nabla \cdot J_i = \sum_j S_{ij} r_j + s_i, \tag{4.6}$$

$$J_i = C_i j - \phi_p D_i \nabla C_i. \tag{4.7}$$

The pore velocity is $v_{\text{rel}} = j/\phi_p$. The chemical switch and the transporting fluid are separately accounted for; their coupling specifies whether switch material moves with that fluid. The homogeneous executed instance has no species boundary flux and uses Equation (4.3). [V2, §9; R4]

5 Thermal rates, enzyme kinetics and biochemical fuel

Source basis V2, §§7–10; S3, p. 12, retains the memory-modulated forward rate.

The temperature-dependent kinetic backbone is retained:

$$k(T) = Ae^{-E_a/(RT)}, \qquad \frac{k(T)}{k(T_r)} = \exp\left[\frac{E_a}{R}\left(\frac{1}{T_r} - \frac{1}{T}\right)\right]. \tag{5.1}$$

A first-order transition-state realization is

$$k(T) = \kappa_{\mathrm{tr}}\frac{k_B T}{h}\exp\left(\frac{\Delta S^{\ddagger}}{R}\right)\exp\left(-\frac{\Delta H^{\ddagger}}{RT}\right) = \kappa_{\mathrm{tr}}\frac{k_B T}{h}e^{-\Delta G^{\ddagger}/(RT)}. \tag{5.2}$$

Activation energies are molar in these expressions. For a higher-order concentration rate law, the standard-state factors and rate-constant units must also be supplied. V3 uses V2's first-order parameterization in its executed instance.

A selected stress/occupancy coupling can enter the barrier through

$$\Delta G^{\ddagger} = \Delta G_0^{\ddagger} + p\Delta V^{\ddagger} + G_c c + G_m m, \qquad \partial_p \ln k = -\frac{\Delta V^{\ddagger}}{RT}. \tag{5.3}$$

Here $\Delta V^{\ddagger}$ is in $\mathrm{m}^3\ \mathrm{mol}^{-1}$; G_c, G_m are in $\mathrm{J\ mol}^{-1}$. This barrier option and the rational rate modulation in §18 are different declared constitutive choices.

Enzyme-mediated conversion

For $E + B_0 \rightleftharpoons EB_0 \to E + B_1$, the inherited quasi-steady initial-rate form is

$$v = \frac{V_{\max}[B_0]}{K_M + [B_0]}, \qquad V_{\max} = k_{\mathrm{cat}}E_T, \qquad K_M = \frac{k_{-1} + k_{\mathrm{cat}}}{k_1}. \tag{5.4}$$

With an explicit reverse pathway, $\dot{c} = (v_{01} - v_{10})/C_\star$. The first-order switch follows only under the corresponding dilute-substrate or pseudo-first-order reduction. [V2, §9]

Lipid–scaffold binding

For $S + L \rightleftharpoons SL$, define $\beta = [SL]/S_T$:

$$\dot{\beta} = k_{\mathrm{on}}[L](1-\beta) - k_{\mathrm{off}}\beta, \qquad \beta_\infty = \frac{[L]}{K_d + [L]}, \quad K_d = \frac{k_{\mathrm{off}}}{k_{\mathrm{on}}}. \tag{5.5}$$

Binding occupancy may modulate swelling, permeability or a rate through a declared function. It is not silently identified with the history occupancy m.

Biochemical fuel and the lever state

The retained ATP-cycle realization

V2 supplied a phosphorylatable switch with reversible channels

$$B_0 + \mathrm{ATP} \rightleftharpoons B_1 + \mathrm{ADP}, \tag{5.6}$$
$$B_1 + \mathrm{H_2O} \rightleftharpoons B_0 + \mathrm{P}_i. \tag{5.7}$$

With chemostatted activities and water absorbed into the indicated coefficient,

$$r_a = k_a^+ [B_0] a_{\mathrm{ATP}} - k_a^- [B_1] a_{\mathrm{ADP}}, \tag{5.8}$$
$$r_b = k_b^+ [B_1] - k_b^- [B_0] a_{\mathrm{Pi}}, \qquad \dot{[B_1]} = r_a - r_b. \tag{5.9}$$

In a closed version, $\dot{[\mathrm{ATP}]} = -r_a$, $\dot{[\mathrm{ADP}]} = r_a$ and $\dot{[\mathrm{P}_i]} = r_b$. Both adenylate and phosphate-group bookkeeping are then explicit:

$$[\mathrm{ATP}] + [\mathrm{ADP}] = \text{constant}, \qquad 3[\mathrm{ATP}] + 2[\mathrm{ADP}] + [\mathrm{P}_i] + [B_1] = \text{constant}. \tag{5.10}$$

Using $\mu_i = \mu_i^\circ + RT \ln a_i$, the forward-cycle affinity is

$$\mathcal{A}_{\mathrm{ATP}} = \mu_{\mathrm{ATP}} + \mu_{\mathrm{H_2O}} - \mu_{\mathrm{ADP}} - \mu_{\mathrm{Pi}}. \tag{5.11}$$

For rates satisfying local detailed balance,

$$\dot{s}_{\mathrm{prod}} = R \sum_j (r_j^+ - r_j^-) \ln(r_j^+ / r_j^-) \geq 0. \tag{5.12}$$

The inequality follows term by term from $(x - y)\ln(x/y) \geq 0$. Fuel turnover, retained state and event count remain separate balances. [V2, §8]

Lipid order and crosshatch coupling

The retained order/scaffold energy density is

$$\begin{aligned} f = {} & W_\ell \ell^2 (1-\ell)^2 + \Delta f(T, c)\ell + \frac{K_\ell}{2} |\nabla \ell|^2 \\ & + \frac{1}{2} (\varepsilon - \varepsilon^*) : \mathbb{C}_q : (\varepsilon - \varepsilon^*). \end{aligned} \tag{5.13}$$

Here $\varepsilon = \operatorname{sym} \nabla u_s$ and $\varepsilon^*(\ell, c, T)$ supplies the swelling/eigenstrain. For nonconserved order, $\dot{\ell} = -L_\ell \ell (1-\ell) \delta F / \delta \ell$, $F = \int f \, \mathrm{d}V$. At fixed other fields and boundaries, $\dot{F} \leq 0$ follows by substitution. A conserved composition instead uses a conservative chemical-potential flux. This module gives the two preparaats a mechanical coupling without replacing their chemical balances. [V2, §10]

6 Clock Alpha, Clock Beta and retained history

Source basis S3, pp. 4–5; V2, §17.

Clock Alpha is the fast active state a; Clock Beta is the slower retained state b. Their inherited equations are

$$\tau_\alpha \dot{a} = -a + \chi b + u, \qquad \tau_\beta \dot{b} = a - b, \qquad \tau_\beta > \tau_\alpha > 0. \tag{6.1}$$

For a held input and $0 \le \chi < 1$,

$$A_{\alpha\beta} = \begin{pmatrix} -1/\tau_\alpha & \chi/\tau_\alpha \\ 1/\tau_\beta & -1/\tau_\beta \end{pmatrix}, \quad a_\infty = b_\infty = \frac{u}{1-\chi}. \tag{6.2}$$

The exact interval update is

$$\begin{pmatrix} a^+ \\ b^+ \end{pmatrix} = \begin{pmatrix} a_\infty \\ b_\infty \end{pmatrix} + e^{A_{\alpha\beta}\Delta t} \left[\begin{pmatrix} a \\ b \end{pmatrix} - \begin{pmatrix} a_\infty \\ b_\infty \end{pmatrix} \right]. \tag{6.3}$$

Its matrix has negative trace and determinant $(1-\chi)/(\tau_\alpha\tau_\beta) > 0$. The fixed-input equilibrium is therefore asymptotically stable. This calculation concerns the held linear flow; feedback changes the sequence of inputs and intervals.

The source's inverse-log retained readout

For committed records $D_1, \dots, D_n$ holding activated fractions c_i, use the V2 normalization

$$w_j = \frac{1}{\ln(j+e)}, \qquad \bar{h}_n = \frac{\sum_{i=1}^{n} w_{n-i} c_i}{\sum_{i=1}^{n} w_{n-i}}, \qquad \bar{h}_0 = 0. \tag{6.4}$$

Positive normalized weights keep $\bar{h}_n$ between the smallest and largest recorded c_i. This readout is a scalar used for evaluation. The archive $\mathcal{H}_n$ retains the full ordered records separately. A digest h_n and numerical readout $\bar{h}_n$ are different objects despite their related letters.

Feedback at the next interval

The executed input is

$$u_n = c_n + g_q q_n + g_{u,h} \bar{h}_n. \tag{6.5}$$

Alpha reads the current fraction, the control bit and the model's own prior outputs. Beta integrates Alpha. The same record readout enters a forward-rate coefficient and a torque coefficient in V3. A record created at event $n+1$ is first used when constructing interval $n+1$, never to alter the interval that produced it.

7 Hourglass dwell, occupancy and recovery

Source basis S3, p. 4; V2, §§18–19; original dialogue S1, pp. 18–23.

Define active dwell and transferred material independently:

$$X(t) = \int_0^t q(s)\,\mathrm{d}s, \qquad M_{\mathrm{tr}}(t) = \int_0^t \int_\Gamma \rho_f j \cdot n\,\mathrm{d}A\,\mathrm{d}s. \tag{7.1}$$

X has units of time; M_{tr} has units of mass. The bounded coordinate $m \in [0,1]$ describes the selected retained occupancy. A constitutive relation specifies how transport fills or empties it.

The cyclic branch retained from V2

$$\dot{m} = \begin{cases} \eta(1-m), & q=1, \\ -(1+m)/\tau_r, & q=0,\ m>0, \\ 0, & q=0,\ m=0. \end{cases} \tag{7.2}$$

The exact held-bit steps are

$$m^+ = \begin{cases} 1-(1-m)e^{-\eta\Delta t}, & q=1, \\ \max\{0,(1+m)e^{-\Delta t/\tau_r}-1\}, & q=0. \end{cases} \tag{7.3}$$

The zero in the return branch is the empty-reservoir boundary of this stated law. It does not freeze the clock, reset the archive or remove a tick. Both branches preserve $[0,1]$.

Starting from $m_0 > 0$ in the return branch, the exact emptying time is

$$T_{\mathrm{rec}} = \tau_r \ln(1+m_0) \le \tau_r \ln 2. \tag{7.4}$$

This recovers the logarithmic recovery expression with an explicit return trajectory.

The inherited one-pass alternative

For a single exposure of length X, the original finite-occupancy branch remains

$$s(X) = \ln(1+\beta_t X), \quad m(X) = 1 - \exp\left[-\frac{\eta_t X}{1+s(X)}\right], \quad T_{\mathrm{rec}}(X) = \tau_r \ln[1+m(X)]. \tag{7.5}$$

This is a different constitutive branch from cyclic filling. V3 executes Equation (7.3), and accumulates $X^+ = X + q\Delta t$. The bit may return to zero while occupancy, elapsed dwell, spin and archive remain advanced. That is the hourglass memory of the completed operation.

8 From the source's record trace to convergence and expansion

Source basis S3, pp. 6–9; V2, §§20–21.

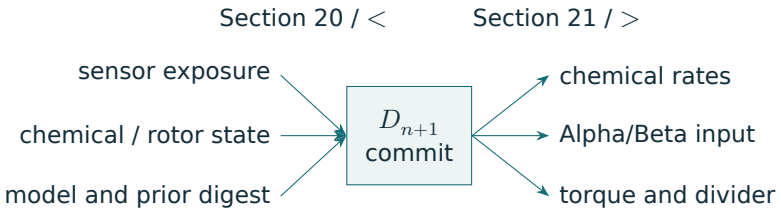
The source traces *cryptographic memory locking* to two operations: Section 20 commits a photographic or state representation; Section 21 reads the committed history to select the next update. It then asks to interpret 20 as $<$ and 21 as $>$.

Preserve those directional roles and define their types:

$$\mathcal{P}_{<} : \text{interval observations and endpoint state} \longrightarrow D_{n+1}, \tag{8.1}$$

$$\mathcal{P}_{>} : (\mathcal{H}_{n+1}, x_{n+1}, q_{n+1}, \mathcal{M}) \longrightarrow \text{next held coefficients.} \tag{8.2}$$

The first operation combines many time-indexed observations into one record. The second distributes information from that record and earlier records into several update channels. Their connection is causal composition, not an inverse transform.



The payload and its address

The model and event identifiers are

$$h_{\mathcal{M}} = \mathrm{SHA256}(\mathrm{enc}(\mathcal{M})), \qquad h_{n+1} = \mathrm{SHA256}(\mathrm{enc}(D_{n+1}^{\mathrm{payload}})). \tag{8.3}$$

The digest has 256 bits, or 32 bytes. The state payload is retained separately and may be much larger. Thus the source's *32-byte pixel address* is formalized here as the digest/address of a record, not as a lossless compression of its entire observations. [S3, pp. 6–9; R2]

A focal event, not a second chemical species

At the junction, Ψ_{dove} denotes the idempotent commitment of the selected representation. In the record-state notation,

$$\Psi_{\mathrm{dove}}(D, \Delta D) = (D, 0), \qquad \Psi_{\mathrm{dove}}^2 = \Psi_{\mathrm{dove}}. \tag{8.4}$$

$\Delta D = 0$ means this particular record is held. The next physical interval continues, and the next event receives its own new record. Sections 20 and 21 give the complete payload and evaluation rules.

9 Double-cone rays and the finite focal interface

Source basis S3, pp. 7–10, including the diagram on p. 9 and the subsequent geometric revision.

V3 construction Separate the ideal convergence/expansion diagram from the finite mechanical axle.

Let $r_\perp = (x, y)$ and choose $L_c > 0$, half-angle $\beta_c \in (0, \pi/2)$. An ideal pair of solid ray envelopes is

$$\mathcal{C}_< = \{(r_\perp, z) : -L_c \le z \le 0,\ |r_\perp| \le (-z)\tan\beta_c\}, \tag{9.1}$$
$$\mathcal{C}_> = \{(r_\perp, z) : 0 \le z \le L_c,\ |r_\perp| \le z\tan\beta_c\}. \tag{9.2}$$

They meet at the ideal vertex 0. With u, v on the appropriate input and output disks, ray segments are

$$\gamma^u_<(s) = ((1-s)u, -(1-s)L_c), \qquad \gamma^v_>(s) = (sv, sL_c), \quad 0 \le s \le 1. \tag{9.3}$$

The two rays have the same junction but opposite convergence behavior. The z coordinate in this schematic is an ordering axis; it need not be the vertical direction of gravity.

The source's "infinitesimally flat" center

For an event record the time slice is

$$\Sigma_{n+1} = \{t = t_{n+1}\}, \qquad D_{n+1} = \mathcal{P}_<\left(x|_{[t_n, t_{n+1}]}, \mathcal{M}, \mathcal{H}_n\right). \tag{9.4}$$

The slice has zero duration in the event description, while its exposure integral covers a finite preceding interval. This is V3's precise meaning of the flat focal checkpoint. It does not identify zero temporal thickness with zero spatial curvature.

Finite mechanical realization

The source's next step replaces a vertex-only picture by a physical diabolo axle. Use a finite throat radius $r_a > 0$ and half-length $z_a > 0$. Its central solid region may be represented by

$$\mathcal{A} = \{(x, y, z) : x^2 + y^2 \le r_a^2,\ |z| \le z_a\}. \tag{9.5}$$

The cups attach at the two rings $z = \pm z_a$, $r = r_a$. The event vertex is an index shared by projection and evaluation; $\mathcal{A}$ is a finite mechanical support. A coupling map places the indexed event at this support without treating the ideal mathematical vertex as an infinite-density physical point.

> The source's double cone supplies the direction of information flow. The open-cup model supplies finite geometry. Their common center is a declared interface, not an instruction to divide by a zero radius.

10 The diabolo: open cups and two external stick-poles

Source basis S3, pp. 10–11. The source photograph on p. 10 shows the open-cup form.

V3 construction A smooth flared profile supplies the geometry that the source leaves qualitative.

For $\sigma \in \{-1, +1\}$, $u \in [0, L_s]$ and $v \in [0, 2\pi)$, define

$$X_\sigma(u, v, t) = x_c(t) + Q_s(t) \begin{pmatrix} r(u) \cos v \\ r(u) \sin v \\ \sigma(z_a + u) \end{pmatrix}, \qquad r(u) > 0. \tag{10.1}$$

A conical instance is $r(u) = r_a + \kappa u$; a flared quadratic instance is $r(u) = r_a + \kappa_1 u + \kappa_2 u^2$. The rims at $u = L_s$ remain open. Neither instance is a pair of closed spherical hemispheres.



Figure 1: Source photograph, S3 p. 10; reproduced from the supplied attachment.

The two poles. Let $s_0(t)$ and $s_H(t)$ be the two external stick-tip positions. Their interface labels remain Γ_0 and Γ_H:

$$U_\Gamma = (s_0, s_H, T_0, T_H, \phi, d).$$

The input and output pole roles refer to these external interfaces. The upper and lower cup rims are not relabeled as the two handsticks.

A realization supplies maps from U_Γ to contact forces, optical alignment and imposed boundary data. The source does not specify stick lengths, friction coefficients or a measured tension series.

Intrinsic shell geometry

The induced metric and area element of Equation (10.1) are

$$g_{uu} = 1 + r'^2, \qquad g_{vv} = r^2, \qquad g_{uv} = 0, \qquad \mathrm{d}A = r\sqrt{1 + r'^2}\,\mathrm{d}u\,\mathrm{d}v. \tag{10.2}$$

For scalar f, the associated surface Laplacian is

$$\Delta_S f = \frac{1}{r\sqrt{1 + r'^2}} \partial_u \left(\frac{r}{\sqrt{1 + r'^2}} \partial_u f \right) + \frac{1}{r^2} \partial_v^2 f. \tag{10.3}$$

A species or order field can evolve on this surface with specified rim and axle conditions. The previous rectangular coordinates survive as local computational charts, not as a compulsory global horseshoe boundary.

11 The string drive, the T operation and topspin

Source basis S3, pp. 11–13.

V3 construction A torque balance makes the source's string drive and topspin explicit; $\mathcal{T}_\pi$ remains the event operator.

The string is a **drive connection**. Its instantaneous shape can transmit forces, but it is not the permanent spatial path along which the archive is indexed. Let the two contact forces about the axle be F_0, F_H at lever arms r_0, r_H. The torque component is

$$\tau_{\text{str}} = (r_0 \times F_0 + r_H \times F_H) \cdot e_z. \tag{11.1}$$

For an ideal tangential drive this reduces to the signed expression $\tau_{\text{str}} = r_a(T_0 - T_H)$, with the tension directions and winding convention fixed by the realization.

Continuous rotational equation

For fixed axial inertia $J > 0$ and drag $\zeta > 0$,

$$J\dot{\omega}_s = \tau_{\text{str}} - \zeta\omega_s - \tau_{\text{load}}, \qquad \dot{\psi} = \omega_s, \qquad L_z = J\omega_s. \tag{11.2}$$

The mechanical energy balance is

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{1}{2}J\omega_s^2\right) = \tau_{\text{str}}\omega_s - \zeta\omega_s^2 - \tau_{\text{load}}\omega_s. \tag{11.3}$$

The selected positive-spin branch is $\omega_s > 0$. It has a declared orientation, angular momentum and work input. A Boolean 1 is not a quantity of torque; $q \oplus 1$ can command a new configuration only through its actuator or selected orientation map. [R3]

Chemical topspin as directional bias

The source prints the forward-rate expression

$$k_{01,n} = k_T \frac{1 + 0.35a_n + 0.25\bar{h}_n}{1 + 0.5m_n}. \tag{11.4}$$

Its accompanying prose describes Alpha as increasing the denominator. The printed equation places Alpha in the *numerator*; V3 retains the equation as the rate definition. Consequently,

$$\frac{\partial k_{01}}{\partial a} = \frac{0.35k_T}{1 + 0.5m} > 0, \qquad \mathcal{A}_{\text{rate}} = \ln\frac{k_{01}}{k_{10}}. \tag{11.5}$$

Positive $\mathcal{A}_{\text{rate}}$ means a forward-biased two-state rate pair; it is not by itself a mechanical torque. V3 connects the bias, the recorded history and the string-driven rotor through the explicit coefficients in §18. This gives the source's topspin both a kinetic coordinate and a mechanical coordinate.

12 Rotation, Coriolis deviation and gravity

Source basis S3, pp. 11 and 13–14.

V3 construction Use the rotating-frame acceleration identity to specify the deviation called Coriolis in the source.

Let the shell rotate with angular velocity $\Omega = \omega_s e_z$. Relative to an origin with acceleration a_O, the inertial and rotating-frame accelerations obey

$$a_I = a_O + a_{\text{rel}} + 2\Omega \times v_{\text{rel}} + \dot{\Omega} \times r + \Omega \times (\Omega \times r). \tag{12.1}$$

All vectors in this identity must be expressed in the same instantaneous basis. The rotating-frame force terms are therefore

$$f_{\text{Cor}} = -2\rho_f \Omega \times v_{\text{rel}}, \tag{12.2}$$
$$f_{\text{cf}} = -\rho_f \Omega \times (\Omega \times r), \tag{12.3}$$
$$f_{\text{Eu}} = -\rho_f \dot{\Omega} \times r. \tag{12.4}$$

These are Coriolis, centrifugal and Euler terms respectively. The shell shape selects tangents, constraints and available flow directions. Coriolis deviation specifically requires nonzero relative motion not parallel to Ω; flared geometry alone does not supply it. [R3]

Coriolis changes direction, not work along the velocity

The cross product gives

$$v_{\text{rel}} \cdot f_{\text{Cor}} = -2\rho_f v_{\text{rel}} \cdot (\Omega \times v_{\text{rel}}) = 0. \tag{12.5}$$

Thus the string or another physical drive supplies the rotational work in §11. Coriolis is the rotating-frame directional term, not a substitute energy source.

Gravity and the source's "G" question

Use g for gravitational acceleration. It acts in the material dynamics and in the laboratory translation balance

$$M\ddot{x}_c = F_0 + F_H + Mg + F_{\text{other}}. \tag{12.6}$$

In the rotating body basis its components are $g_B = Q_s^T g_I$. An axially aligned gravity vector remains constant in that basis; a differently oriented one need not.

The source retains gravity while identifying the string-as-permanent-boundary as the red herring. V3 keeps that distinction. The term g enters force and transport balances; the identity $P_1 + P_2 = P_\star$ follows from the selected moiety balance. Setting $g = 0$ changes a force term, not the algebra in Equation (4.5).

Rotating porous transport and its shell projection

The inherited nonrotating branch

With permeability tensor K in m^2, fluid viscosity μ_f in Pa s and density ρ_f,

$$j = -\frac{K}{\mu_f}(\nabla p - \rho_f g). \tag{12.7}$$

This is the Darcy flux convention retained from V2. It is accompanied by a continuity equation and chosen pressure or flux conditions at the material ports. Gravity can contribute to the driving potential. [V2, §18; R4]

A declared rotating-frame closure

For a quasi-steady linear-drag extension, constant porosity $\phi_p > 0$ and pore velocity $v_{\mathrm{rel}} = j/\phi_p$, define

$$b_{\mathrm{eff}} = g - a_O - \dot{\Omega} \times r - \Omega \times (\Omega \times r). \tag{12.8}$$

Keeping the Coriolis term while neglecting the remaining relative acceleration yields the selected closure

$$\left[\mu_f K^{-1} + \frac{2\rho_f}{\phi_p}[\Omega\times]\right] j = -\nabla p + \rho_f b_{\mathrm{eff}}. \tag{12.9}$$

$[\Omega\times]$ is the skew-symmetric cross-product matrix. Equation (12.9) combines the inherited Darcy drag convention with the rotating-frame identity; it is a V3 closure, not an equation printed in S3.

If K is symmetric positive definite, then for every nonzero y,

$$y^T\left[\mu_f K^{-1} + \frac{2\rho_f}{\phi_p}[\Omega\times]\right] y = \mu_f y^T K^{-1} y > 0. \tag{12.10}$$

The skew term contributes no quadratic work, and the displayed coefficient matrix is nonsingular. A specified pressure/body-force field therefore defines a unique local flux under this closure.

Tangential flow on the open cup

For shell unit normal n_S, let $P_T = I - n_S n_S^T$ and $\nabla_S p = P_T \nabla p$. The nonrotating tangential branch is

$$j_S = -\frac{K_S}{\mu_f}(\nabla_S p - \rho_f P_T g), \qquad \partial_t(\phi_p C_i) + \nabla_S \cdot J_{i,S} = \sum_j S_{ij} r_j + s_i. \tag{12.11}$$

Here K_S is positive definite on the two-dimensional tangent plane. Rim, axle and contact boundary conditions complete a surface field realization. The reduced 48-tick run evaluates a sampled orientation-dependent conductance, not this spatial pressure problem.

13 Crosshatch tensors and the quarter-turn exchange

Source basis S3, pp. 12 and 15; V2, §§10 and 18.

The source's crosshatch rotation is retained:

$$K_q = R_q K_0 R_q^T, \qquad \sigma_q = R_q \sigma_0 R_q^T, \qquad R_q = R_z(\pi q/2). \tag{13.1}$$

For fourth-order elasticity the corresponding rule is

$$(\mathbb{C}_q)_{ijkl} = R_{ia} R_{jb} R_{kc} R_{ld} (\mathbb{C}_0)_{abcd}, \tag{13.2}$$

with repeated indices summed. Permeability K and stress σ are not interchangeable tensors: K_{xy} has units m^2, while σ_{xy} has units Pa.

For a planar $K_0 = \mathrm{diag}(k_\parallel, k_\perp)$ and laboratory angle $\alpha = \psi + \pi q/2$,

$$K(\alpha) = \begin{pmatrix} k_\parallel \cos^2\alpha + k_\perp \sin^2\alpha & (k_\parallel - k_\perp)\sin\alpha\cos\alpha \\ (k_\parallel - k_\perp)\sin\alpha\cos\alpha & k_\parallel \sin^2\alpha + k_\perp \cos^2\alpha \end{pmatrix}. \tag{13.3}$$

At fixed ψ, changing q exchanges K_{xx} and K_{yy} and reverses K_{xy}. The eigenvalues, trace and determinant remain the same. At principal-axis alignment the off-diagonal component is zero, even though the two conductance directions can differ.

The crosshatch lever

A selected poroelastic stress law is

$$\sigma = \mathbb{C}_q : [\varepsilon(u_s) - \varepsilon^*(c, \ell, T)] - \alpha_B p I, \qquad \nabla \cdot \sigma + f = 0. \tag{13.4}$$

For a member direction n, a stress-sensitive pore radius can be specified as

$$r_p = r_{p0} e^{-\beta_p \sigma_c}, \qquad \sigma_c = -n^T \sigma n \quad \text{for tension-positive stress.} \tag{13.5}$$

The coefficient β_p has inverse-pressure units. This constitutive step connects state, stress, pore size and drainage. Rotation alone changes components relative to external directions; it does not create new stress eigenvalues or a shear energy without a loading/deformation law.

Two mechanical descriptions

The reduced model assigns the selected tensor immediately from (ψ, q). A resolved crosshatch actuator instead evolves its orientation, for example

$$J_\varphi \ddot{\varphi} + \zeta_\varphi \dot{\varphi} = -\partial_\varphi U + \tau_{\mathrm{act}}, \qquad U = B \sin^2(2\varphi) - h_c \cos(2\varphi). \tag{13.6}$$

The tick then specifies the commanded well. V2's actuator and V3's sampled tensor are explicit realization choices, not an unmodeled mechanical impulse.

14 The whaletail: flared paths, flux and directional routing

Source basis S3, pp. 14–15: the whaletail, invisible arch and $\mathcal{T}_\pi$ as the rudder.

V3 construction The source gives a shape correspondence; V3 supplies a parametric two-wing envelope and a conservative flux construction.

Use a planar chart with the focal interface at the origin and terminal projections at $(\pm L_w, 0)$. A simple symmetric pair of wing centerlines is

$$\gamma_\pm(s) = \big(\pm L_w s,\ A_w \sin(\pi s)\big), \qquad 0 \le s \le 1, \quad L_w, A_w > 0. \tag{14.1}$$

Each branch flares away from the focus and returns to its terminal level. The source does not provide a unique curvature law; Equation (14.1) is the declared V3 envelope.

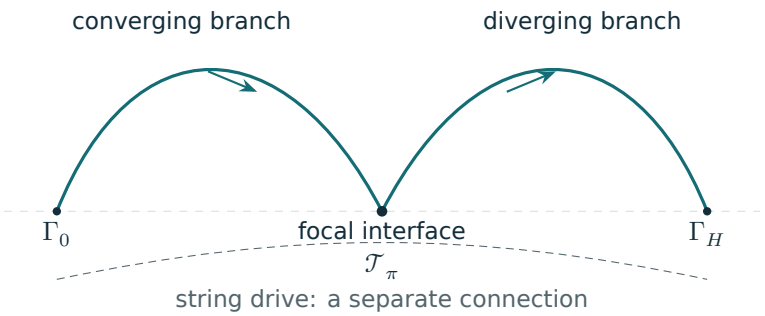


Figure 2: Schematic whaletail envelope. The source's two terminal roles and central T operation are retained; the curve is a V3 construction, not a measured flow image.

A two-dimensional ribbon, not the string centerline

Let $n_\pm(s)$ be unit normals to the wing curves. Define finite-width ribbons

$$F_\pm(s,\eta) = \gamma_\pm(s) + \eta w(s) n_\pm(s), \quad -\tfrac{1}{2} \le \eta \le \tfrac{1}{2}, \quad w(s) > 0. \tag{14.2}$$

Choose width and curvature so the Jacobian $\mathcal{J} = |\det(\partial_s F, \partial_\eta F)|$ is positive. Their two-dimensional neighborhoods represent the flared transport chart. The physical string instead enters through its forces and torque.

One branch can carry input from Γ_0 toward the focus and the other output from the focus toward Γ_H. The centerlines are parametrized outward from the focus on both sides; the input branch therefore uses the opposite flux sign. An actual species, optical or magnetic realization supplies its own field equations and interface conditions. Calling all of these possibilities "flux" does not identify their physical conservation laws.

The whaletail flux law and the T rudder

A divergence-free ribbon construction

For constant signed discharge $Q_{\pm}$ per unit out-of-plane depth, define contravariant flux components on each ribbon by

$$\mathcal{J}^s = \frac{Q_{\pm}}{\mathcal{J}}, \qquad \mathcal{J}^\eta = 0, \qquad j = \mathcal{J}^s \partial_s F + \mathcal{J}^\eta \partial_\eta F. \tag{14.3}$$

Then

$$\nabla \cdot j = \frac{1}{\mathcal{J}} \left[\partial_s (\mathcal{J}\mathcal{J}^s) + \partial_\eta (\mathcal{J}\mathcal{J}^\eta) \right] = 0. \tag{14.4}$$

With the outward-from-focus convention, a no-storage junction satisfies $Q_- + Q_+ = 0$. A junction reservoir instead has $\dot{M}_J = -\rho_f (Q_- + Q_+)$ per unit depth. This is a constructed conservative field on the chosen whaletail ribbons. It does not determine the pressure or prove that an unspecified air-gap medium realizes that field.

Directional weights for the executable architecture

The reduced model uses the tensor of §13 to define two nonnegative routing weights:

$$G_n = w_{0,n} = \frac{K_{xx}(\psi_n, q_n)}{k_\parallel + k_\perp}, \qquad w_{H,n} = \frac{K_{yy}(\psi_n, q_n)}{k_\parallel + k_\perp}. \tag{14.5}$$

Consequently,

$$w_0 + w_H = 1, \qquad w_0(\psi, 1-q) = w_H(\psi, q). \tag{14.6}$$

For positive $k_\parallel, k_\perp$, both lie strictly between zero and one. They are directional *coefficient weights*, distinct from the signed physical-port discharges above. The source's $\mathcal{T}_\pi$-rudder correspondence is thereby explicit: at fixed spin angle, the quarter-turn exchanges the two directional weights.

The forward rate in the V3 instance reads G_n through a coefficient g_G:

$$k_{01,n} = k_T \frac{1 + g_a a_n + g_h \bar{h}_n + g_G G_n}{1 + g_m m_n}. \tag{14.7}$$

This is the added geometry-to-kinetics map. The source supplies the whaletail/tensor association, while the additive normalized coefficient is a V3 choice. Setting $g_G = 0$ restores V2's chemical-clock trajectory while leaving the new geometry available as an output.

> The whaletail is not the string. The string drives the rotor; the rotor and bit orient the tensor; the tensor selects directional weights; the selected weights enter the next rate; the resulting event is recorded and read back.

15 Slit, tilt-shift, OTAN2 and thermal exposure

Source basis S3, pp. 7 and 10–11; V2, §§4–5 and 12.

For aperture coordinates $(\xi, \eta) = R(-\gamma)(x - x_0, y - y_0)$, retain the finite slit

$$A_{w,h,\gamma} = \mathbf{1}_{|\xi| \le w/2} \mathbf{1}_{|\eta| \le h/2}, \qquad \mathcal{S}E = A_{w,h,\gamma} E. \tag{15.1}$$

The source-named vertical slit is the selected $w \ll h$ aperture. The optical map is

$$E_{\text{out}} = \mathcal{O}_{\phi,d} \mathcal{S} E_{\text{in}}, \qquad \tilde{x}' \sim H_{\phi,d} \tilde{x} \tag{15.2}$$

for a supplied plane-to-plane homography. Optical propagation, projective coordinate mapping and logarithmic address encoding are distinct operations.

The regularized focal address

Retain the architecture-specific name OTAN2 and define

$$\rho_\epsilon = \frac{M}{2} \ln\left(\frac{x^2 + y^2 + \epsilon^2}{r_0^2} \right), \qquad \bar{\rho} = \frac{M}{2} \ln\left(1 + \frac{x^2 + y^2}{\epsilon^2} \right). \tag{15.3}$$

$M > 0$ is dimensionless; $r_0, \epsilon > 0$ are lengths. At the origin,

$$\rho_\epsilon(0,0) = M \ln(\epsilon / r_0), \qquad \bar{\rho}(0,0) = 0. \tag{15.4}$$

For a nonzero vector, store its quadrant-correct bearing θ_{xy}. At the origin use the convention $(\theta_{xy}, v_\theta) = (0, 0)$, where v_θ marks that a unique bearing is absent. A nonzero radius uses $v_\theta = 1$. The finite logarithm and the physical throat radius are separate parameters unless a calibration explicitly connects them.

Octant implementation retained from V2

Set $a_x = |x|$, $a_y = |y|$ and $z = \min(a_x, a_y) / \max(a_x, a_y)$ for nonzero input. An approximation $P(z) \approx \arctan z$ gives

$$\alpha = \begin{cases} P(z), & a_x \ge a_y, \\ \pi/2 - P(z), & a_y > a_x, \end{cases} \qquad \theta_{xy} = \begin{cases} \alpha, & x \ge 0, y \ge 0, \\ \pi - \alpha, & x < 0, y \ge 0, \\ \alpha - \pi, & x < 0, y < 0, \\ -\alpha, & x \ge 0, y < 0. \end{cases} \tag{15.5}$$

The approximation, zero convention and target's selection instructions belong to its implementation. V3 does not reinterpret the source term as the name of a separately supplied software function. The event address can combine $(\bar{\rho}, \theta_{xy}, v_\theta, N, q)$ while keeping spatial bearing distinct from tick phase.

Thermal, photonic and depth-clock equations

Two-sided exposure along a material chart

For an opposed path coordinate $z \in [0, H]$, retain

$$\mu_a(c) = \mu_1(1-c) + \mu_2 c, \qquad \partial_z I_+ = -\mu_a I_+, \qquad \partial_z I_- = +\mu_a I_-. \tag{15.6}$$

The signs follow the two propagation directions. The source's high-absorption or Vanta-state term denotes large optical depth in this model. The absorption coefficients remain supplied material parameters. [V2, §12]

With enthalpy h_v per bulk volume, thermal conductivity κ_T, molar reaction enthalpy ΔH_j and rates per bulk volume,

$$\partial_t h_v(T, \ell) = \nabla \cdot (\kappa_T \nabla T) + \mu_a(I_+ + I_-) - \sum_j \Delta H_j r_j + Q_{\text{ext}}. \tag{15.7}$$

A latent-heat contribution included in h_v is not included again as an independent source. A surface-shell instance substitutes its appropriate surface heat balance and boundary exchange law.

Local chemical progress

The inherited one-way exposure branch is

$$\partial_t c = k_f(T)\frac{I_+ + I_-}{I_\star}(1-c), \qquad \vartheta_{\text{chem}}(z,t) = \int_0^t k_f(T)\frac{I_+ + I_-}{I_\star}\,\mathrm{d}s. \tag{15.8}$$

At fixed location, $c = 1 - [1 - c(0)]e^{-\vartheta_{\text{chem}}}$. Surface and interior can therefore accumulate different chemical progress during the same elapsed atomic-reference time.

For constant temperature and absorption, one-sided $I = I_0 e^{-\mu z}$, $c(z,0) = 0$ and a chosen activation threshold $0 < c_\star < 1$,

$$z_f(t) = \frac{1}{\mu}\ln\!\left(\frac{k_f I_0 t}{I_\star[-\ln(1-c_\star)]}\right), \qquad \dot{z}_f = \frac{1}{\mu t}. \tag{15.9}$$

The expression is restricted to times after the surface reaches the threshold and before the front leaves the modeled depth. It is a logarithmic penetration branch with inverse-time front speed. If μ_a changes with conversion, the coupled equations above determine the front instead. [V2, §12]

Connection to commitment

The sensor or material response is integrated over the held interval and passed to Section 20. The exposure integral is finite even though its committed event is indexed by one endpoint. A committed image or record holds a representation of the interval; it is not an instruction to set the continuing material temperature or every chemical rate to zero.

16 Atomic reference, Rabi inversion and Ramsey feedback

Source basis V2, §§13–15; S3, pp. 3 and 5, retains the paced clock and reference-clock connection.

An atomic transition supplies an energy/frequency relation

$$\Delta E = E_e - E_g = h\nu_A = \hbar\omega_A, \qquad \omega_A = 2\pi\nu_A. \tag{16.1}$$

The retained ideal reference uses the defining caesium-133 frequency

$$\nu_{\mathrm{ref}} = \Delta\nu_{\mathrm{Cs}} = 9\,192\,631\,770\ \mathrm{Hz}. \tag{16.2}$$

The SI second is defined through the unperturbed ground-state hyperfine transition. The biochemical rates and rotor frequency are separate quantities. [R1]

Carrier phase and counted time

For a reference oscillator,

$$\Phi_A(t) = \Phi_A(0) + 2\pi \int_0^t \nu_{\mathrm{ref}}(s)\,\mathrm{d}s, \qquad C_A(t) = \frac{\Phi_A(t) - \Phi_A(0)}{2\pi}. \tag{16.3}$$

At the selected fixed reference, $t = C_A/\nu_{\mathrm{ref}}$. The executed clock stores integer full-cycle counts and chooses a divider on each interval. It does not equate one reference cycle with one molecular jump.

The constants retained from V2 are $h = 6.62607015 \times 10^{-34}$ J s, $k_B = 1.380649 \times 10^{-23}$ J K^{-1} and $R = 8.31446261815324$ J mol^{-1} K^{-1}. They provide the units linking the chemical and atomic equations. [V2, §13]

A driven atom as a bit realization

For a two-level atom in the rotating-wave approximation, use

$$H_R = \frac{\hbar}{2}\left[\delta\sigma_z + \Omega_R(t)(\cos\phi_d\,\sigma_x + \sin\phi_d\,\sigma_y)\right], \qquad \dot\rho = -\frac{i}{\hbar}[H_R, \rho]. \tag{16.4}$$

Here δ and Ω_R are angular frequencies. At resonance with a fixed x-axis drive,

$$\vartheta_R = \int \Omega_R(t)\,\mathrm{d}t, \qquad U(\vartheta_R) = e^{-i\vartheta_R\sigma_x/2}, \qquad U(\pi) = -i\sigma_x. \tag{16.5}$$

Thus $U(\pi)|q\rangle = -i|1-q\rangle$: the populations are exchanged, up to a global phase. Multiplying a state vector by -1 without this driven rotation does not exchange populations. The Rabi pulse supplies the physical inversion map; the architecture supplies its event label. [V2, §14; R5]

Pulse completion, Ramsey discrimination and clock roles

For constant drive and an initial ground state, the two-level excited-state probability is

$$P_e(t) = \frac{\Omega_R^2}{\Omega_R^2 + \delta^2} \sin^2\left(\frac{\sqrt{\Omega_R^2 + \delta^2}\, t}{2}\right). \tag{16.6}$$

On resonance, $t_\pi = \pi/\Omega_R$. For the selected example $\Omega_R/(2\pi) = 1$ kHz, $t_\pi = 0.5$ ms. This pulse time is set by the drive coupling, not by half the caesium carrier period. [V2, §14; R5]

The retained Ramsey reference loop

In the ideal discriminator convention used in V2,

$$P_e(\phi) = \frac{1}{2}\left[1 + C_R \cos(\delta T_R + \phi)\right], \quad e = P_e(-\pi/2) - P_e(+\pi/2) = C_R \sin(\delta T_R). \tag{16.7}$$

An illustrative integral frequency correction is

$$\delta_{j+1} = \delta_j - \frac{g_R}{T_R} e_j. \tag{16.8}$$

Near the central fringe, $\delta_{j+1} \simeq (1 - g_R C_R)\delta_j$; the local linear condition is $0 < g_R C_R < 2$. The Ramsey index j labels reference interrogation cycles. It need not equal the architectural event index n. [V2, §15]

Which clock does what?

Coordinate	Function	Connection to the event
Φ_A	Counts reference carrier phase	Integer cycles determine Δt_n.
ϑ_R	Rotates atomic populations under drive	A completed π pulse can realize a toggle.
ϑ_{chem}	Accumulates reaction progress	The chemical-rate law can select pacing.
a, b	Fast and retained state response	Sampled state and history change later rates.
ψ	Mechanical spin angle	Rotates the directional conductance.
Θ	Counts architectural half-turn events	Increases by π at each completed tick.

The V3 numerical execution uses an ideal fixed carrier and the held biochemical divider, not a numerical Schrödinger or Ramsey-servo integration. A driven-atom implementation can instead report completed π pulses through the same event interface. The same event algebra therefore supports distinct physical timing realizations without identifying their frequencies.

17 Clock division and the unified flow-and-tick map

Source basis V2, §§16 and 22; S3, pp. 4–5.

For a positive intended event rate r_n held on interval n, choose

$$D_n^{\text{cycles}} = \left\lceil \frac{\nu_{\text{ref}}}{r_n} \right\rceil, \qquad \Delta t_n = \frac{D_n^{\text{cycles}}}{\nu_{\text{ref}}}, \qquad C_{A,n+1} = C_{A,n} + D_n^{\text{cycles}}. \tag{17.1}$$

The time and architectural phase over that interval are

$$t_n = \frac{C_{A,n}}{\nu_{\text{ref}}}, \qquad \Theta(t) = n\pi + \frac{\Phi_A(t) - \Phi_A(t_n)}{2D_n^{\text{cycles}}}. \tag{17.2}$$

Exactly D_n^{cycles} carrier cycles give $\Delta\Phi_A = 2\pi D_n^{\text{cycles}}$, hence $\Delta\Theta = \pi$. No additional phase jump is added after this endpoint accounting: $\Theta_{n+1} = (n+1)\pi$ is the event value of the same accumulated phase.

The ceiling construction gives the interval quantization identity

$$\frac{1}{r_n} \le \Delta t_n < \frac{1}{r_n} + \frac{1}{\nu_{\text{ref}}}. \tag{17.3}$$

It bounds the rounding of a selected held interval, not the accuracy of the chosen chemical or material coefficients.

The hybrid map

Let x contain the continuous coordinates of the selected realization and λ_n its held rates, inputs, torque and geometry. Define

$$x_{n+1} = \text{Flow}_{\mathcal{M}}(\Delta t_n; x_n, q_n, \lambda_n), \tag{17.4}$$

$$(N_{n+1}, q_{n+1}, \Theta_{n+1}) = (N_n + 1, 1 - q_n, \Theta_n + \pi). \tag{17.5}$$

Section 20 then commits the event and Section 21 evaluates the next coefficient set. The full composition is

$$\mathcal{S}_{n+1} = \text{Commit}_{<} \circ \mathcal{T}_{\pi} \circ \text{Flow}_{\mathcal{M},\lambda_n}(\mathcal{S}_n), \qquad \lambda_n = \text{Eval}_{>}(\mathcal{M}, x_n, q_n, \mathcal{H}_n). \tag{17.6}$$

The tick does not automatically jump concentrations, occupancy or rotor angular velocity. Those quantities follow their flow law unless an explicit impulsive realization has been selected. In the reduced model, the newly toggled bit changes the geometry and branch read at the next interval start.

> Continuous flow determines the endpoint state. The event updates parity and count. Commitment records what happened. Evaluation uses that record to determine what happens next.

18 Fully specified V3 held-interval instance

Source basis V2, §23, supplies the chemical, Alpha/Beta and hourglass instance.

V3 construction Add a continuous rotor, sampled anisotropic routing and a history-dependent drive.

The executed continuous state is $x = (c, a, b, m, X, \psi, \omega_s)$. Set $T = T_r = 298.15$ K, $E_a = 50\,000$ J mol^{-1} and $k_r = 1$ s^{-1}:

$$k_T = k_r \exp\left[\frac{E_a}{R}\left(\frac{1}{T_r} - \frac{1}{T}\right)\right]. \tag{18.1}$$

At the start of each interval, read $\bar{h}_n$ from Equation (6.4) and G_n from Equation (14.5). The complete rate pair is

$$k_{01,n} = k_T \frac{1 + 0.35a_n + 0.25\bar{h}_n + 0.15G_n}{1 + 0.5m_n}, \tag{18.2}$$

$$k_{10,n} = 0.6k_T \frac{1 + 0.2b_n}{1 + 0.5m_n}, \tag{18.3}$$

$$r_n = (1 - q_n)k_{01,n} + q_n k_{10,n}. \tag{18.4}$$

The divider uses Equation (17.1). The fraction uses the exact held-rate solution (4.3).

Alpha/Beta and memory

Use $\tau_\alpha = 0.2$ s, $\tau_\beta = 4$ s, $\chi = 0.2$ and

$$u_n = c_n + 0.2q_n + 0.2\bar{h}_n. \tag{18.5}$$

The dual-clock state uses Equation (6.3). Memory uses $\eta = 0.45$ s^{-1}, $\tau_r = 5$ s in Equation (7.3); dwell is updated by $X_{n+1} = X_n + q_n \Delta t_n$.

Anisotropic rotor coupling

For the routing ratio, set normalized $k_\parallel = 4$ and $k_\perp = 1$. Only their ratio is used; these are not measured permeabilities. The held drive is

$$\tau_n = \tau_0(1 + 0.3a_n + 0.2\bar{h}_n), \qquad \tau_0 = 2 \times 10^{-4}\ \text{N m}. \tag{18.6}$$

Set $J = 2 \times 10^{-4}$ kg m^2, $\zeta = 4 \times 10^{-5}$ N m s and $\tau_{\text{load}} = 0$. The routing tensor is sampled at (ψ_n, q_n); the exact rotor flow evolves ψ and ω_s during the interval.

Initialize

$$(c, a, b, m, X, \psi, \omega_s)_0 = (0.1, 0, 0, 0, 0, 0, 2), \qquad q_0 = N_0 = C_{A,0} = 0. \tag{18.7}$$

The archive is initially empty. These declared coefficients completely specify the reduced execution. Full shell shape, optical exposure fields, ATP-pool dynamics and spatial Coriolis transport are not additional hidden states of this reduced instance.

Exact rotor flow and event execution order

For one held torque, write $\lambda = \zeta/J$ and $\omega_\infty = \tau_n/\zeta$. The exact step is

$$\omega_{s,n+1} = \omega_\infty + (\omega_{s,n} - \omega_\infty)e^{-\lambda \Delta t_n}, \tag{18.8}$$

$$\psi_{n+1} = \psi_n + \omega_\infty \Delta t_n + \frac{\omega_{s,n} - \omega_\infty}{\lambda}(1 - e^{-\lambda \Delta t_n}). \tag{18.9}$$

For positive initial spin and nonnegative held drive, ω_s stays positive. It is a convex combination of its initial value and the torque-dependent equilibrium. This mechanical positivity does not follow from bit parity.

Independent mechanical energy residual

Set $d = \omega_{s,n} - \omega_\infty$. The analytic interval integrals are

$$I_1 = \int \omega_s \, \mathrm{d}t = \omega_\infty \Delta t + \frac{d}{\lambda}(1 - e^{-\lambda \Delta t}), \tag{18.10}$$

$$I_2 = \int \omega_s^2 \, \mathrm{d}t = \omega_\infty^2 \Delta t + \frac{2\omega_\infty d}{\lambda}(1 - e^{-\lambda \Delta t}) + \frac{d^2}{2\lambda}(1 - e^{-2\lambda \Delta t}). \tag{18.11}$$

The offline residual calculation checks

$$\epsilon_E = \frac{1}{2}J(\omega_{s,n+1}^2 - \omega_{s,n}^2) - \tau_n I_1 + \zeta I_2. \tag{18.12}$$

It vanishes in exact arithmetic by the rotor equation. The finite-precision maximum from the actual run is reported in §19.

One interval, in order

1. Read the already committed history and compute $\bar{h}_n$.
2. Evaluate $K(\psi_n, q_n)$, G_n, both rates, u_n and τ_n.
3. Select integer D_n^{cycles} and Δt_n.
4. Apply the exact chemical, Alpha/Beta, memory and rotor flows; accumulate dwell and carrier cycles.
5. Apply $\mathcal{T}_\pi$ to the event count and bit, and set the endpoint phase count.
6. Commit the new state with the model identifier, prior digest and held coefficients. The next interval reads this newly extended history.

All rates and driving coefficients are held from the interval start. In particular, G_n is not repeatedly re-sampled while the rotor moves during that interval. This timing convention makes the reduced calculation reproducible and identifies exactly where geometry feeds into the chemistry.

19 Executed trajectory, comparisons and identities

The reference program was executed for 48 ticks with the parameters in §18. It was run again from the same model; its complete canonical record sequence matched. The following values are extracted from the generated event records. The routing column is the weight held on the interval ending at event n.

n	t_n (s)	q_n	c_n	m_n	$\omega_{s,n}$	held G
1	0.892857	1	0.532496	0.000000	2.490607	0.800000
2	2.554196	0	0.672319	0.526500	3.393487	0.690935
3	3.369078	1	0.700737	0.296931	3.927457	0.576044
4	5.145779	0	0.701113	0.683938	4.830882	0.340151
8	10.298476	0	0.697583	0.764692	6.385723	0.734069
16	20.401168	0	0.691912	0.765416	7.225094	0.791060
24	30.393798	0	0.685336	0.768534	7.347353	0.480200
32	40.436548	0	0.689607	0.759736	7.372424	0.693225
40	50.393701	0	0.685494	0.768450	7.369228	0.492013
48	60.434427	0	0.690368	0.759458	7.376330	0.734500

At event 48,

$$\Theta_{48} = 48\pi, \qquad q_{48} = 0, \qquad t_{48} = 60.434427445 \text{ s}, \tag{19.1}$$

$$\psi_{48} = 412.592715764 \text{ rad}, \qquad \omega_{s,48} = 7.376329801 \text{ rad s}^{-1}, \tag{19.2}$$

$$X_{48} = 40.667188190 \text{ s}, \qquad C_{A,48} = 555551437734. \tag{19.3}$$

The bit has completed 24 full two-tick cycles. The independently driven rotor has completed approximately 65.666170 mechanical turns. The event phase and rotor angle have different values because they are connected by the declared timing and orientation laws, not identified.

Explicit comparison runs

Instance	t_{48} (s)	c_{48}	m_{48}
V3, full feedback	60.434427445	0.690368	0.759458
History coefficients zero	63.709681333	0.656973	0.741154
Mechanical spin zero	60.102146351	0.679753	0.770512
V2 projection	61.205110590	0.674976	0.754033

The history-off run sets the coefficients of $\bar{h}$ in the forward rate, Alpha input and torque to zero, leaving the archive present. The spin-off run sets initial ω_s and τ_0 to zero; binary orientation still alternates. The V2 projection sets only the newly added rate coefficient g_G to zero.

The full run minus the history-off completion time is -3.275253888 s. This is the calculated effect of the selected coefficients. The V2 projection reproduces all 48 stored V2 values for the 18 compared inherited fields, including carrier cycles, rates, input, fraction, Alpha/Beta, occupancy, dwell and time.

Derived invariants and executed residuals

Algebraic identities

Parity. Induction on $q^+ = q \oplus 1$ gives $q_n = q_0 \oplus (n \bmod 2)$. The phase counter stores $\Theta_n/\pi = n$ as an integer, rather than a rounded large-angle product.

Carrier endpoint. Each interval accumulates D_n^{cycles} full reference cycles. Equation (17.1) therefore gives exactly one architectural half-turn per event by definition.

Population and memory. The exact chemical update is a convex combination. Both selected hourglass branches preserve $[0, 1]$. The normalized positive history weights preserve the same range for $\bar{h}$.

Tensor exchange. Substituting $\alpha + \pi/2$ into Equation (13.3) exchanges its diagonal components and reverses its off-diagonal component. Equation (14.6) follows without changing eigenvalues.

Causal record readout. The coefficient for interval n uses only $D_1, \ldots, D_n$. The event at its end is D_{n+1}. There is no dependency on an unavailable future record.

Results of the executed checks

Check	Executed result
Tick index, parity and integer phase count	Pass
Divider and accumulated carrier-cycle account	Pass
All recorded c, m inside $[0, 1]$	Pass
Readout recomputed from each preceding record prefix	Exact match
Model identifier and predecessor digest chain	Pass
Second canonical execution from the same model	Exact match
V2 projection, 48 events $\times$ 18 inherited fields	Exact match
Maximum $\lvert(1 - c) + c - 1\rvert$	0
Maximum $\lvert w_0 + w_H - 1\rvert$	2.221×10^{-16}
Maximum rotor energy residual $\lvert\epsilon_E\rvert$	1.735×10^{-18} J
Quarter-turn routing exchange at held spin angle	Pass
Positive rotor speed in the selected full run	Pass

The residuals are arithmetic evaluations of the stated model. Record comparisons use the same canonical encoding and execution environment. They establish the reported computational identities and replay, not a measured material tolerance.

> V3's loop is active in the reference implementation: its own records change later rates and drive; drive changes spin; spin and bit change routing; routing changes the next interval; the new event is committed under the same rule schema.

20 Photographic commitment and model-bearing records: $<$

Source basis S3, pp. 6–9; V2, §20. This section retains the source's Section 20 identity.

An exposure record starts from

$$E_{\text{exp}}(x) = \int_{t_n}^{t_{n+1}} I_{\text{sensor}}(x,t)\,\mathrm{d}t, \qquad D_{\text{image}} = \mathcal{Q}\mathcal{R}(E_{\text{exp}}). \tag{20.1}$$

$\mathcal{R}$ is the selected response and $\mathcal{Q}$ its encoder. In the reduced execution the observed object is the model state rather than an image field. The endpoint record remains the same kind of time-indexed commitment.

Finite self-description

Define the model snapshot as

$$\mathcal{M} = (\text{rule schema}, \text{parameters}, \text{initial state}, \text{source digests}, \text{implementation digest}). \tag{20.2}$$

Its canonical digest $h_{\mathcal{M}}$ is stored in every event. The snapshot exists once; events refer to it rather than embedding recursively repeated copies.

The emitted payload is

$$\begin{aligned} D_{n+1}^{\text{payload}} = (&h_{\mathcal{M}}, h_n, N_{n+1}, q_n, q_{n+1}, C_{A,n+1}, D_n^{\text{cycles}},\\ &t_{n+1}, \Delta t_n, x_{n+1}, \bar{h}_n, u_n, k_{01,n}, k_{10,n}, \tau_n, K_n, w_{0,n}, w_{H,n}). \end{aligned} \tag{20.3}$$

The implementation also records its integer Θ/π count, intended rate and rotor-energy residual. It computes

$$h_{n+1} = \text{SHA256}(\text{enc}(D_{n+1}^{\text{payload}})), \qquad \mathcal{H}_{n+1} = \mathcal{H}_n \| (D_{n+1}^{\text{payload}}, h_{n+1}), \qquad h_0 = h_{\mathcal{M}}. \tag{20.4}$$

The hash excludes its own digest field. That exclusion and the predecessor index make the dependency graph finite and ordered.

Petrification at the interface

$\Psi_{\text{dove}}(D, \Delta D) = (D, 0)$ holds this record's values. Reapplying it gives the same held record. New observations produce a new event identifier rather than revising the committed endpoint. The source's $<$ is the convergence of state, exposure, rule identity and prior history into this one payload.

Meaning of cryptographic memory locking

In V3 this source phrase names *record commitment plus digest linkage*. A recomputed digest can be compared with the retained one to test byte-level identity. A hash does not itself preserve storage, establish authorship or prevent an entire alternative chain from being recomputed; those are different properties from the stated equality check. The payload, 32-byte digest and history storage remain separate parts of the architecture. [R2]

21 Causal self-reference under the same equations: $>$

Source basis S3, pp. 5–9 and 15; V2, §21. This section retains the source's Section 21 identity.

Section 21 is the evaluation fan-out from the committed record into the next set of coefficients. Write $\mathcal{M} = (F, G, \mathcal{R}_H, \mathcal{C}; \lambda)$ for the finite flow rule, output rule, readout, coefficient selector and parameter set.

The complete causal recursion is

$$\bar{h}_n = \mathcal{R}_H(\mathcal{H}_n; \mathcal{M}), \tag{21.1}$$
$$\lambda_n = \mathcal{C}(x_n, q_n, \bar{h}_n; \mathcal{M}), \tag{21.2}$$
$$\Delta t_n = D_n^{\text{cycles}} / \nu_{\text{ref}}, \tag{21.3}$$
$$x_{n+1} = \text{Flow}_{F,\mathcal{M}}(\Delta t_n; x_n, q_n, \lambda_n), \tag{21.4}$$
$$(N_{n+1}, q_{n+1}, \Theta_{n+1}) = (N_n + 1, 1 - q_n, \Theta_n + \pi), \tag{21.5}$$
$$D_{n+1} = G(\mathcal{M}, \mathcal{H}_n, x_{n+1}, q_n, q_{n+1}, \lambda_n), \tag{21.6}$$
$$\mathcal{H}_{n+1} = \mathcal{H}_n \| D_{n+1}. \tag{21.7}$$

In V3, λ_n contains the chemical rates, Alpha input, selected torque, sampled conductance and integer divider. The next invocation reads the archive that this invocation has extended.

The source's expanding cone

The same retained information enters different channels:

$$\bar{h}_n \longmapsto \begin{pmatrix} k_{01,n} \\ u_n \\ \tau_n \end{pmatrix} \longmapsto \begin{pmatrix} \Delta t_n \\ (a_{n+1}, b_{n+1}) \\ (\psi_{n+1}, \omega_{s,n+1}) \end{pmatrix}. \tag{21.8}$$

This is the $>$ fan-out in equations. The hash locates and identifies the retained payload; the numerical readout comes from its recorded c_i values, not by trying to invert SHA-256.

Self-reference is a dependency, not merely a label

An output is operationally self-referential here when it is read by the same model's later state update. The reduced program reads a finite rule schema and parameters, emits their identifier and state, then reads its own prior emitted fractions. Section 19's history-off comparison changes those coefficients while leaving the archive present, separating an operative feedback path from passive record keeping.

Because evaluation is delayed to the next interval, the recursion requires no self-containing event object or instantaneous fixed-point solution. The source's two opposing operations meet at one indexed event and continue through time.

Document self-reference and the complete V3 identity

What this document refers to

This edition names its source attachment, carries forward V2's biochemical and clock rules, defines its added geometry and rotor maps, publishes the executable parameter set, and includes the result of evaluating that set. Its source, implementation and model digests are recorded in Appendix D. The numerical event sequence points back to the same model and implementation that produced it.

The PDF is the committed specification and report. The reference evaluator, model snapshot and evolving archive are its active realization. This assignment preserves the source's photographic commitment and recursive evaluation without treating a static page as an autonomously running physical process.

An acyclic self-reference construction

The actual dependency order is

$$\begin{aligned}\text{source files and program} &\longrightarrow \text{model snapshot} \longrightarrow h_{\mathcal{M}} \\ &\longrightarrow D_1 \longrightarrow D_2 \longrightarrow \cdots \longrightarrow D_{48} \longrightarrow \text{report.}\end{aligned} \tag{21.9}$$

The report can publish the model and event identifiers because those objects were computed first. Its own final PDF hash belongs in an external manifest; it is not inserted into the same bytes whose digest it is meant to equal.

The full geometry-to-history cycle

Preparaat 1 supplies the structured role. Preparaat 2 supplies the active state. Their biochemical balance evolves c. The hourglass retains dwell in m and X. Alpha and Beta carry fast and slow response. The two stick-poles impose the selected drive and alignment. The string transfers torque; it is not the stored whaletail route. Mechanical spin and the binary quarter-turn orient K. The whaletail weighting changes the selected rate. Reference cycles pace the next π endpoint. Section 20 commits the event; Section 21 reads it back.

In operator form,

$$\boxed{\begin{aligned}\lambda_n &= \mathcal{P}_{>}(\mathcal{M}, x_n, q_n, \mathcal{H}_n), \\ \mathcal{S}_{n+1} &= \mathcal{P}_{<} \circ \mathcal{T}_{\pi} \circ \mathrm{Flow}_{\lambda_n}(\mathcal{S}_n), \\ \Delta\Theta = \pi, \qquad q^{+} &= q \oplus 1, \qquad \mathcal{H}^{+} = \mathcal{H} \| D^{+}.\end{aligned}} \tag{21.10}$$

V3 architectural identity
Convergence records the operation. The T event changes the bit. Expansion evaluates the recorded operation. The next operation therefore carries the history of the same architecture that produced it.

A Source concordance and edition decisions

The primary base is *wait I got 2 preparaats.pdf*, 15 pages, abbreviated S3. Page numbers below refer to that attachment, not this report. V2 is the prior 34-page *Tom_Klootwijk_Architectural_Paradigm_v2_Pi_Clock.pdf*. S1 is the earlier 49-page *trewel.pdf*.

Source element	S3 pages	V3 treatment	Provenance of equations
Preparaat 1: structured capacity	2	Capacity/scaffold role; §1	Source role and inherited partition.
Preparaat 2: activated process	3	$P_2 = P_\star c$; §§1, 4	Source equation; V2 switch kinetics.
Five-step closure	4–5	Ordered flow, tick, commit, read; §§2, 17	Source sequence; explicit hybrid map.
Double Hadamard	4	Masked co-location; §1	Source product; normalized form derived.
Alpha/Beta and hourglass	4	§§6–7	Source clock equations; V2 exact flow/recovery.
Record/model hash trace	6–7	§§8, 20	Source record lineage; explicit implementation digest added.
20 as $<$, 21 as $>$	7–9	§§8–9, 20–21	Source directions; typed maps and ray coordinates added.
Open, non-globular cups	10	Finite shell and axle; §§9–10	Source geometry; parametric profile added.
String and Coriolis	11	Drive and rotating-frame terms; §§11–12	Source role; torque/acceleration completion.
Sticks as true poles	11, 13	External Γ_0, Γ_H; §10	Source assignment retained.
T and topspin	11–13	Bit operation, rate bias and spin; §§2, 11	Source terminology; separate quantities defined.
Gravity / red herring	13–14	Force term versus routing; §12	Source gravity retained; moiety identity separated.
Whaletail / invisible arch	14–15	Two-wing ribbon and weights; §14	Source shape correspondence; V3 construction.
Quarter-turn tensor rudder	15	§§13–14	Source K_q; explicit component exchange derived.

Named but not supplied as a separate source. S3 page 2 names *militair-cologne.pdf*. Its contents are not among the files available for this edition. V3 uses the Preparaat 1 description actually present in S3; it does not invent the contents of the named document. The opening medical disambiguation on S3 pages 1–2 is not an architectural module.

Source language and explicit mathematical resolution

The following entries retain the source wording as terminology and identify the precise interpretation used in the formal model. They also make visible the places where a new definition or a distinct equation is needed.

Source statement or phrase	Exact V3 reading
"32-byte pixel address"; S3 p. 9	The 32-byte object is h_D, a digest. The full record D is stored separately. A digest is not an invertible encoding of the observation window.
"Infinitesimally flat" focal vertex; pp. 7–9	Zero-duration event index $t = t_n$ in the record map. The mechanical realization uses a finite axle region with $r_a, z_a > 0$.
Epsilon prevents collapse at the axle; p. 11	$\epsilon > 0$ makes the log-radius finite. Mechanical support comes from the specified geometry and force law, not from adding a number inside a logarithm.
Alpha increases the forward-rate denominator; p. 12	The equation printed on that page places $0.35a_n$ in the numerator. V3 retains the printed formula and derives its positive numerator sensitivity.
Coriolis from rapid flared-shell rotation; p. 11	The Coriolis term is $-2\rho_f \Omega \times v_{\text{rel}}$. Shape selects the permissible relative flow; it does not replace the velocity factor.
K_{xy} as shear stress; pp. 11–12	K_{xy} is a permeability/conductance component. Shear stress is σ_{xy}. A constitutive force/strain law relates mechanics to transport.
Gravity needed for $P_1 + P_2$ balance; p. 13	Gravity appears in force/transport equations. The closed switch population is conserved by opposite reaction terms or $l^T S = 0$, including the $g = 0$ case.
String "vanishes" from core transformations; p. 14	String shape is excluded as the archive's permanent route. The transmitted torque still enters the rotor equation through the declared drive port.
Whaletail is the literal flux profile; pp. 14–15	The source supplies the geometric association. V3 declares one two-wing envelope, a conservative ribbon flux and a separate reduced routing law. Boundary data select any material-field solution.
"String theory"; pp. 14–15	The attachment uses this phrase for a mistaken one-dimensional string-path interpretation. V3 does not introduce high-energy string-theory equations.

The document's proposed architecture, source terminology and mathematical implications remain separately traceable. The executed branch is the coefficient set and update order in §18, rather than every alternative continuum realization at once.

B Complete parameters and initial conditions

All values below are the input values of the executed 48-tick instance. Inherited numerical coefficients come from V2; rotor and routing coefficients are the selected V3 extension.

Parameter	Value	Unit	Role
ν_{ref}	9 192 631 770	Hz	Ideal reference
T, T_r	298.15	K	Temperature and reference
E_a	50 000	J mol^{-1}	Arrhenius activation energy
k_r	1	s^{-1}	Reference forward scale
Reverse ratio	0.6	1	Relative reverse scale
τ_α	0.2	s	Alpha response time
τ_β	4	s	Beta response time
χ	0.2	1	Beta-to-Alpha coupling
g_a, g_b	0.35, 0.2	1	Rate-state coefficients
g_h, g_m	0.25, 0.5	1	Rate history and occupancy
g_G	0.15	1	Added routing coefficient
$g_q, g_{u,h}$	0.2, 0.2	1	Held Alpha input
η	0.45	s^{-1}	Memory filling
τ_r	5	s	Recovery scale
$k_\parallel, k_\perp$	4, 1	normalized	Conductance ratio
J	2×10^{-4}	kg m^2	Rotor inertia
ζ	4×10^{-5}	N m s	Linear rotational drag
τ_0	2×10^{-4}	N m	Held-drive scale
$g_{\tau,a}, g_{\tau,h}$	0.3, 0.2	1	Torque feedback coefficients
Tick count	48	integer	Run length

Initial conditions are $c_0 = 0.1$, $a_0 = b_0 = m_0 = X_0 = \psi_0 = 0$, $\omega_{s,0} = 2$ rad s^{-1}, $q_0 = N_0 = C_{A,0} = 0$ and $\mathcal{H}_0 = ()$.

How to reproduce the execution

From the companion package directory, using Python 3.9 or later:

```
python whaletail_clock.py --model model.json --out results
```

The program uses only the Python standard library. It writes `events.csv`, `events.json`, `checks.json` and `model_snapshot.json`. The separately supplied `v2_projection_check.json` records the comparison with the original V2 event file. The report's tables are generated from these event outputs rather than transcribed from the source dialogue.

C Executable kernel and representation contract

The full source program is in the companion package. The following excerpt gives the exact scalar chemistry, rate, timing and memory part of the V3 loop. The named dual_clock and rotor_step routines implement the analytic flows in §§6 and 18.

```
hbar = history_readout(records)
geom = routing(state["psi"], q, p)
denominator = 1 + p["g_memory"] * m
k01 = kt * (1 + p["g_alpha"] * a
             + p["g_history"] * hbar
             + p["g_transport"] * geom["w0"]) / denominator
k10 = kt * p["reverse_ratio"] * (1 + p["g_beta"] * b) / denominator
rate = k01 if q == 0 else k10
divider = math.ceil(p["nu_ref"] / rate)
dt = divider / p["nu_ref"]
ceq = k01 / (k01 + k10)
cn = ceq + (c - ceq) * math.exp(-(k01 + k10) * dt)
u = c + p["input_q"] * q + p["input_history"] * hbar
an, bn = dual_clock(a, b, u, dt, p)
mn = (1 - (1 - m) * math.exp(-p["fill_rate"] * dt) if q == 1
      else max(0.0, (1 + m) * math.exp(-dt / p["tau_recovery"]) - 1))
torque = p["torque_base"] * (1 + p["torque_alpha"] * a
                             + p["torque_history"] * hbar)
wn, psin, energy_residual = rotor_step(
    state["omega"], state["psi"], torque, dt, p)
cycles += divider
# Update the state, complement q, form the event payload,
# then hash that payload and append it to records.
```

Canonical record encoding

The reference encoder uses JSON with sorted keys, compact separators, ASCII escaping, no NaN values and UTF-8 bytes. Its digest field is excluded from its own input. Numerical arithmetic is Python binary64 float arithmetic with the platform math library; integer event and carrier counts remain integers. The reported replay was obtained using Python 3.13.5 on Linux.

Phase and geometry representation

The field `theta_over_pi` is the integer event index. The field `psi` is the separate accumulated rotor angle in radians. The routing routine reduces that angle modulo 2π for trigonometric evaluation, then applies the $q\pi/2$ orientation. This avoids mixing the event phase count with the rotating laboratory conductance.

D Source, implementation and event identities

The digests below were computed from the actual supplied files and emitted records. File and program entries are SHA-256 over file bytes. The model entry hashes the canonical model snapshot. The final event entry hashes its canonical payload excluding its own digest field. Each displayed pair of lines forms one continuous 64-character hexadecimal digest.

wait I got 2 preparaats.pdf

```
3a5f902a537c76d476870f5e1c70c8f1
51ecab12d3060bb0bbfd4533328a341a
```

Tom_Klootwijk_Architectural_Paradigm_v2_Pi_Clock.pdf

```
687bc20c54f8242e2d9a00755d35f169
196ca8c1ab752e85752fd68462ca2298
```

whaletail_clock.py

```
59bff0013aa43ac7bc49849101d823da
f79b19a443259b0a79e060187998c78a
```

canonical model snapshot

```
72bd6a406b3e611021144ef2476bd33f
193b30cf72f21a9635cea81a634b0efa
```

event 48

```
2cd16cb24c2814415c0ef95091050024
3c75809a86b4e3e3c1c2121757cb1c61
```

Artifact dependency and attribution

The supplied author attribution is Tom Klootwijk, NL200678942 / 10-07-1990. The source-document digests identify the exact attachment and V2 file used in this edition. The implementation digest identifies the reference program included with it. These data establish the report's reproducible input and output identities; they do not substitute a digest for the underlying documents.

Final-file manifest

The companion package includes an external manifest for the delivered PDF, program, model and result files. The PDF's final byte digest is calculated after typesetting. Keeping that digest outside the PDF avoids a circular requirement for a file to contain the hash of its own final bytes.

E References and source basis

Supplied source documents

S3 Tom Klootwijk, supplied dialogue *wait I got 2 preparaats.pdf*, 15 pages. Primary base of this edition. Source roles, equations, questions and later geometry revisions are mapped in Appendix A.

V2 *The Klootwijk Architectural Paradigm: Biochemical and atomic clocks with executable self-reference*, Version 2.0, 16 October 2026, 34 pages. Attributed to Tom Klootwijk. Supplied in this conversation together with *Klootwijk_Pi_Clock_Reference_Package.zip*. Source of the retained biochemical, atomic, dual-clock, hourglass and divider equations and the baseline event data.

S1 Tom Klootwijk, supplied dialogue *trewel.pdf*, 49 pages. Earlier source of OTAN2, two-sided exposure, slit/tilt-shift, crosshatch, hourglass and Alpha/Beta terminology. Used through the explicit inherited equations and source map of V2.

P Tom Klootwijk, preceding instruction in this conversation: add biochemical and atomic-clock equations, make the architecture self-referential, and interpret a tick as a one-bit flip over π.

Primary and official references used for V3's completion

The external references below support the named mathematical/physical conventions. The cross-domain coupling coefficients and geometric profile functions remain the V3 definitions identified in the text. Sources accessed 16 October 2026.

R1 Bureau International des Poids et Mesures, *SI base unit: second (s)*. Official definition fixing the unperturbed caesium-133 frequency at 9 192 631 770 Hz.

R2 National Institute of Standards and Technology, *Secure Hash Standard (SHS), FIPS PUB 180-4*, 2015, especially §6.2 for SHA-256. Message digests and byte-change detection; the record-chain construction itself is specified in §§20–21.

R3 J. Peraire and S. Widnall, MIT OpenCourseWare, *16.07 Dynamics, Lecture L8: Relative Motion using Rotating Axes*, Fall 2009. Rotating-frame derivative and acceleration transformations, especially pp. 3–5.

R4 COMSOL, version 6.4 documentation, *Darcy's Law—Equation Formulation*. Darcy flux, permeability/viscosity units, continuity and gravity convention. The Coriolis-extended closure in §12 is an explicitly declared V3 combination.

R5 MIT OpenCourseWare, *5.61 Physical Chemistry, Lecture 36: Time Dependence of Two-Level Systems: Density Matrix, Rotating Wave Approximation*, Fall 2017, pp. 6–8. Driven two-state populations and π-pulse inversion. The retained Ramsey discriminator is stated explicitly in V2 §15 and this edition §16.

End of Version 3.0. The canonical model, executed records, source concordance and event equations together form this edition's self-referential specification.